

# TANVIRAJ JAYAM

9741375332 — tanvi2k5@gmail.com — linkedin — github

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## EDUCATION

### PES University

*Computer Science, Bachelor of Technology*

Bengaluru, India

Aug. 2023 – May 2027

- GPA- 9.28
- Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming ,Database Technologies , Machine Learning , Data Analytics , Big Data, Operating Systems, Software Engineering

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## EXPERIENCE

### Software Developer Intern

*iTuring.Ai*

June 2026 – July 2026

Bengaluru, India

- Kubernetes Infrastructure & Deployment Automation
  - \* Deployed a multi-node Kubernetes cluster with a private Docker Registry and shared NFS storage.
  - \* Automated the deployment of 20+ applications using Helm, accelerating cluster provisioning.
- ML Model Drift Optimization Framework
  - \* Developed a framework to mitigate data drift in ML models by using L-BFGS, CMA-ES, and MoE .
  - \* Built a unified optimization pipeline supporting 3 production ML model families (GBM, Random Forest, GLM) across H2O, scikit-learn, and PMML formats, enabling efficient adaptation of diverse production models.

### Mentee

*Pure Storage*

March 2026 – June 2026

Bengaluru, India

- Developed an eBPF (libbpf + CO-RE) based Linux daemon to monitor and attribute Docker container Out-of-Memory kills in real time using kernel tracepoints.
- Integrated Prometheus and Grafana to visualize container memory metrics with zero-overhead host monitoring.

### Research Intern

*Centre for Data Science and Applied Machine Learning, PES University*

June 2025 – July 2025

Bengaluru, India

- Developed a multilingual hate speech detection system, with a 2% improvement in accuracy (reaching 84%) by fine-tuning transformer-based models on Kannada-English code-mixed datasets.

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## PROJECTS

### MiniRAFT Consensus System | *Node.js, WebSockets, Docker* Apr 2026

- Engineered a distributed consensus system using a simplified RAFT protocol to maintain consistent state across multiple nodes.
- Implemented leader election, heartbeat-based failure detection, majority-based log replication, and automatic leader recovery for fault-tolerant operation.
- Designed a Dockerized multi-node architecture with WebSocket-based inter-node communication and real-time state synchronization.

### AI Driven OS level Environment manager | *MCP server, DistillBERT*

Jan 2026 – Present

- Building an Agentic AI system to convert natural language into OS commands using a multi-agent pipeline.
- Implementing Containerd-based sandboxed execution with error handling and feedback-driven correction.

### Legal Case Similarity Analyzer | *FAISS, LegalBERT*

March 2026

- Built a hybrid Retrieval-Augmented Generation (RAG) pipeline combining dense retrieval (FAISS ANN indexing) with sparse retrieval (BM25) to improve search relevance and retrieval accuracy.
- Fine-tuned LegalBERT for domain-specific understanding and integrated LLaMA (versatile) as a post-retrieval validation layer to enhance result reliability and reduce irrelevant outputs.

### Dynamic Content Stream with Kafka | *Apache Kafka, Flask, MySQL*

Nov 2025

- Engineered a dynamic Kafka-based streaming system enabling automated topic creation and real-time data flow by integrating Kafka Admin APIs and multi-threaded producers in data-driven content platforms.
- Developed a centralized metadata database and interactive web interface by linking Kafka consumers with a control layer to allow live topic management and user-topic visualization in real-time streaming environments.

## TECHNICAL SKILLS

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**Languages:** Python, C/C++, JavaScript, HTML/CSS

**Frameworks:** Tensorflow/Keras , PyTorch, Streamlit

**Databases:** MySQL, MongoDB

**Developer Tools :** Linux, Git, Github, Docker, Kubernetes, Helm, Jira

## ACCOMPLISHMENTS

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- Hackathon Winner, PES University- Top in 150+ teams; Awarded Rs 20,000 for innovation.
- Scholarship University topper(CNR)- awarded 40% tuition waiver for academic excellence.
- KCET College topper- secured highest rank in the college for Karnataka's state entrance exam.
- Mentored students and lead technical clubs and teams.